

Manipulating Light–Matter Interactions in Plasmonic Nanoparticle Lattices

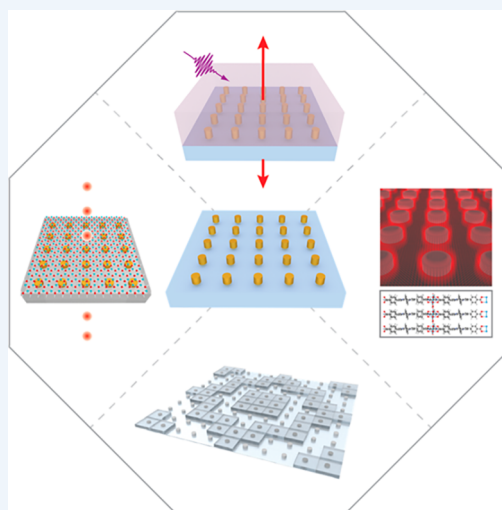
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Danqing Wang,[†] Jun Guan,[†] Jingtian Hu,[‡] Marc R. Bourgeois,[§] and Teri W. Odom^{*,†,‡,§}

[†]Graduate Program in Applied Physics, [‡]Department of Materials Science and Engineering, and [§]Department of Chemistry, Northwestern University, Evanston, Illinois 60208, United States

CONSPECTUS: Rationally assembled nanostructures exhibit distinct physical and chemical properties beyond their individual units. Developments in nanofabrication techniques have enabled the patterning of a wide range of nanomaterial designs over macroscale ($> \text{in.}^2$) areas. Periodic metal nanostructures show long-range diffractive interactions when the lattice spacing is close to the wavelength of the incident light. The collective coupling between metal nanoparticles in a lattice introduces sharp and intense plasmonic surface lattice resonances, in contrast to the broad localized resonances from single nanoparticles. Plasmonic nanoparticle lattices exhibit strongly enhanced optical fields within the subwavelength vicinity of the nanoparticle unit cells that are 2 orders of magnitude higher than that of individual units. These intense electromagnetic fields can manipulate nanoscale processes such as photocatalysis, optical spectroscopy, nonlinear optics, and light harvesting.

This Account focuses on advances in exciton–plasmon coupling and light–matter interactions with plasmonic nanoparticle lattices. First, we introduce the fundamentals of ultrasharp surface lattice resonances; these resonances arise from the coupling of the localized surface plasmons of a nanoparticle to the diffraction mode from the lattice. Second, we discuss how integrating dye molecules with plasmonic nanoparticle lattices can result in an architecture for nanoscale lasing at room temperature. The lasing emission wavelength can be tuned in real time by adjusting the refractive index environment or varying the lattice spacing. Third, we describe how manipulating either the shape of the unit cell or the lattice geometry can control the lasing emission properties. Low-symmetry plasmonic nanoparticle lattices can show polarization-dependent lasing responses, and multiscale plasmonic superlattices—finite patches of nanoparticles grouped into microscale arrays—can support multiple plasmon resonances for controlled multimodal nanolasing. Fourth, we discuss how the assembly of photoactive emitters on the nanocavity arrays behaves as a hybrid materials system with enhanced exciton–plasmon coupling. Positioning metal–organic framework materials around nanoparticles produces mixed photon modes with strongly enhanced photoluminescence at wavelengths determined by the lattice. Deterministic coupling of quantum emitters in two-dimensional materials to plasmonic lattices leads to preserved single-photon emission and reduced decay lifetimes. Finally, we highlight emerging applications of nanoparticle lattices from compact, fully reconfigurable imaging devices to solid-state emitter structures. Plasmonic nanoparticle lattices are a versatile, scalable platform for tunable flat optics, nontrivial topological photonics, and modified chemical reactivities.



1. INTRODUCTION

1.1. Overview

The rational assembly of metal nanoparticles (NPs) into artificial solids with two-dimensional (2D) and three-dimensional (3D) architectures enables unprecedented properties not accessible in their individual constituents.^{1,2} For example, 3D NP superlattices from bottom-up DNA-mediated assembly can achieve tailored optical permittivity, electronic transport, and photoconductivity.^{3–5} Top-down lithography and direct laser writing methods can realize a wide range of metamaterials and metasurfaces composed of subwavelength meta-atoms.⁶

Engineered 3D metamaterials exhibit electromagnetic responses and optical properties distinct from those of natural materials, such as optical cloaking, negative refractive index, and perfect absorption.^{7,8} Compared with complex building units in these artificially structured materials, 2D metasurfaces with simple planar units can be produced by scalable fabrication methods and directly integrated with optoelectronic devices.⁸ Most demonstrations of metasurfaces, however, rely

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on selective control over the phase or polarization of individual components to manipulate the optical wavefront and propagation of light into the far field; there is no collective coupling between units.^{1,9}

2D lattices of metal NPs can support diffractive interactions between units when the particle–particle spacing is close to the wavelength of the incident light.^{10–12} Surface plasmons are collective oscillations of conduction electrons confined to a dielectric–metal interface.¹³ Individual metal NPs support localized surface plasmons (LSPs) with broad resonance line widths (full width at half-maximum (fwhm) > 100 nm);¹⁴ however, NPs in an array surrounded by a uniform dielectric environment exhibit ultrasharp plasmonic surface lattice resonances (SLRs, or lattice plasmons) (fwhm < 5 nm).^{15–17} SLRs show a characteristic Fano line shape from coupling between the LSP and a diffraction mode defined by the lattice, and the in-phase oscillations among NPs introduce suppressed radiative loss.^{15,18} The resonance wavelength can be tuned by changing the refractive index of the environment, the NP size, the interparticle spacing, or the lattice geometry.^{19–21} The strongly enhanced and localized electromagnetic fields (intensity ratio $|E|^2/|E_0|^2 > 10^3$) near the subwavelength vicinity of the NPs can be exploited in a diverse range of applications such as lasing, modified photoluminescence, and nonlinear optics.^{22–25}

This Account focuses on a hybrid materials platform based on plasmonic NP lattices and photoactive materials for controlling light–matter interactions on the nanoscale. First, we introduce the principles of obtaining ultrasharp SLR resonances in plasmonic NP lattices. Control over lattice designs from either unit cell NP shape or lattice symmetry enables access to different SLR wavelength regimes for targeting different chemical processes (Figure 1a). We discuss

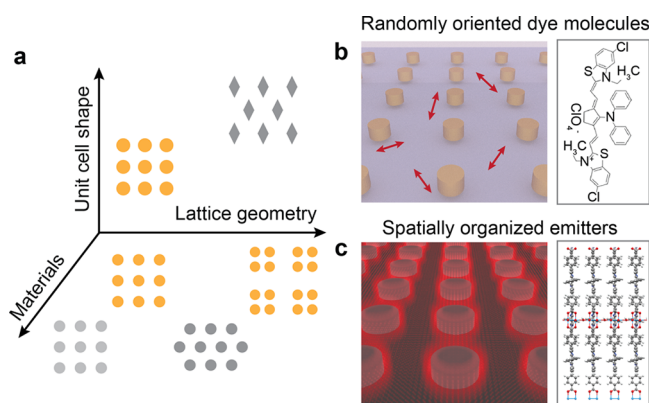


Figure 1. Overview of light–matter manipulation in plasmonic nanoparticle (NP) lattices based on rationally engineered lattices and spatially organized emitters. (a) Scheme of the structural and materials engineering in plasmonic lattices, from square and hexagonal lattices to low-symmetry rhombohedral lattices and multiscale superlattices in later sections. (b, c) Coupling of plasmonic NP lattices with (b) randomly positioned dye molecules and (c) spatially organized molecular emitters.

how integrating NP lattices with dye molecules can achieve coherent light sources at the nanoscale at room temperature, where the emission wavelength is tunable by varying the refractive index and lattice spacing (Figure 1b). Next, we describe how spatially organized photoactive materials assembled around plasmon nanocavities can lead to nanoscale

exciton–plasmon coupling and mixed photon modes (Figure 1c). Finally, we highlight how scalable plasmonic NP lattices are a promising model system for reconfigurable metalenses, topological photonics, and cavity-modified chemical reactions.

1.2. Ultrasharp Resonances in Plasmonic Nanoparticle Lattices

Figure 2 summarizes characteristics of narrow SLR resonances in plasmonic NP lattices.^{10,12,16} Cylindrical NPs with programmable lattice spacing can be generated over square centimeter areas²⁶ using large-scale nanofabrication tools (Figure 2a); macroscale samples enable direct measurements of angle-resolved transmission spectra that reveal optical band structures. Figure 2b shows the zeroth-order transmission spectra of SLRs in Ag NP lattices with narrow line widths (fwhm $\approx$ 3 nm). Varying the lattice spacing a_0 and the corresponding NP diameter d can tune the SLR wavelength over the visible to near-infrared regime. Historically, diffractive coupling between plasmonic NPs was first analyzed using the coupled dipole approximation (CDA) method, which treats each NP as an electric dipole and accounts for dipolar electromagnetic coupling between NPs.¹⁵ Compared with early experimental reports of relatively broad SLRs,²⁷ sharper resonances (line widths < 10 nm) were observed later as the sample uniformity improved and collection angles were narrowed.^{10,16,17} The CDA method can predict the position of the SLR wavelength, where the real parts of α_{NP}^{-1} (where α_{NP} is the NP polarizability) and the dipole lattice sum S are equal to each other.^{17,22} Extinction efficiency spectra for 2D Ag NP lattices calculated using the CDA method¹⁵ agree well with experiments (Figure 2c). Although simple analytic expressions for dipole polarizabilities exist only for spherical (or spheroidal) NPs, numerical multipole expansion²¹ can be employed to provide insight into the mechanisms of mode coupling between more complicated units. For cases where the CDA method is inadequate, finite-difference time-domain (FDTD) simulations can model the time- and position-dependent electromagnetic fields and incorporate arbitrary unit cell shapes, complex lattice designs, and inhomogeneous index environments.²⁸

The angle-resolved optical band structure of plasmonic NP lattices suggests that light can be trapped at SLR band-edge modes (in-plane wavevector $k_{||} = 0$) with a nearly zero group velocity ($v = \delta\omega/\delta k_{||}$). Propagating SLR modes at nonzero wavevectors follow the $(\pm 1, 0)$ diffraction modes of the lattice under transverse electric (TE) polarization and $(0, \pm 1)$ diffraction modes under transverse magnetic (TM) polarization (Figure 2d). Cold-cavity lifetime measurements can characterize the ultrafast photon dynamics within an optical cavity and provide insight on intrinsic losses of band-edge SLRs.²⁹ Pumped by a femtosecond laser near the SLR wavelength at normal incidence, Au and Ag NP lattices showed photon lifetimes τ of 12 and 8 ps, respectively, while Ti NPs exhibited no resolvable τ (Figure 2e). Compared with the 10 fs lifetimes of LSPs in small NPs,¹³ plasmonic NP lattices at the SLR resonance exhibit surprisingly long (ps) lifetimes because of long-range (mm) photon scattering within the pump excitation area.³⁰ Such photon dynamics indicates that SLRs can offer optical feedback in an open-cavity system for enhanced optical processes.

1.3. Enhanced Nonlinear Optics from Plasmonic Lattices

Besides the linear properties, SLRs can boost nonlinear optical responses by their intense near-field electromagnetic enhance-

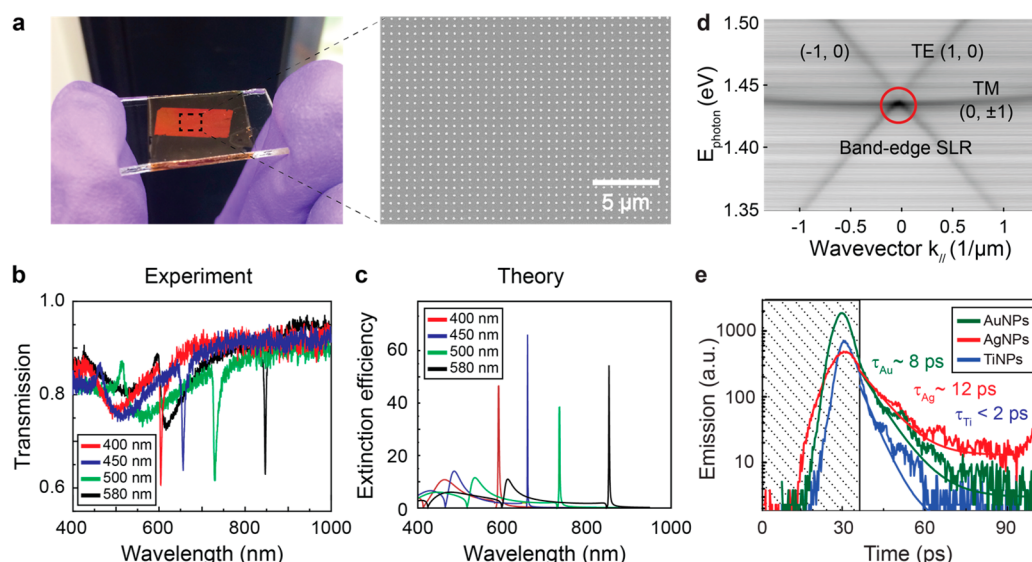


Figure 2. Ultrasharp surface lattice resonances (SLRs) in metal nanoparticle lattices. (a) Photograph and scanning electron microscopy (SEM) image of fabricated Au NP lattices. (b) Experimental demonstration of sharp SLRs spanning the visible and near-infrared regimes with varying lattice spacing a_0 and NP diameter d in Ag NPs. d varies from 80 nm to 130 nm for a_0 from 450 to 600 nm. Transmission spectra were measured at normal incidence. Fused silica was used as the substrate and dimethyl sulfoxide (DMSO) as the superstrate to create a uniform-index environment (refractive index $n = 1.46$) around the NPs. (c) Extinction efficiency (ratio of extinction to geometric cross section) spectra of 2D Ag NP lattices with varying spacings calculated using the coupled dipole method. Adapted with permission from ref 15. Copyright 2004 American Institute of Physics. (d) Dispersion diagram of SLRs. Both transverse electric (TE) and transverse magnetic (TM) modes appear under illumination with unpolarized white light. (e) Ultrafast histogram of photon counts at SLR wavelengths. Adapted with permission from ref 30. Copyright 2019 American Chemical Society.

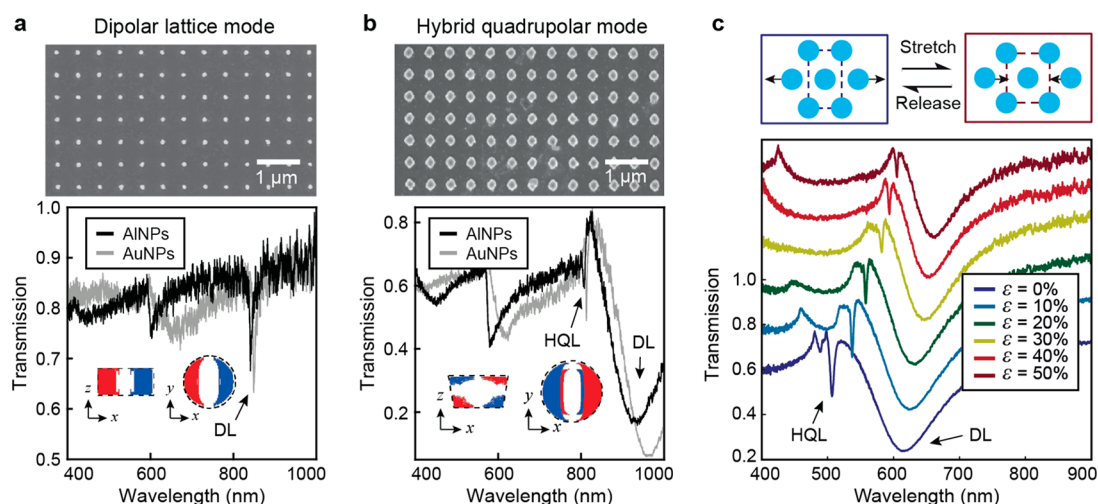


Figure 3. Arrays of Al NPs allow sharp SLR excitations in the visible regime. (a, b) (top) SEM images, (middle) transmission spectra (in comparison with Au), and (bottom) charge distributions of Al NP arrays with (a) dipolar and (b) hybrid quadrupolar SLR excitations. Adapted with permission from ref 37. Copyright 2019 The Optical Society. The cylinder NPs had $d = 110$ nm and $h = 50$ nm in (a) and $d = 220$ nm and $h = 120$ nm in (b). Narrow resonances were also observed at around 580 nm due to a $(1, 1)$ diffraction mode. (c) Tuning of SLR resonance by variation of the NP array spacing. The array in (c) consisted of NPs with $d = 180$ nm and $h = 50$ nm with a spacing of $a_0 = 346$ nm increased in the x direction to 520 nm. Adapted from ref 19. Published by the National Academy of Sciences.

ments. For example, plasmonic NP lattices^{23,31,32} can produce strong second harmonic generation (SHG), a nonlinear optical process that enables shorter-wavelength light to be generated through frequency doubling,³³ as well as unique characterization methods for chemical reactions³⁴ and biological processes.³⁵ Pumped with a near-infrared laser at normal incidence, the narrow-band-edge SLR produced strong SHG signals at ultraviolet wavelengths ($\lambda = 390$ nm) and up to 450-fold enhancement relative to off-resonance signals. Other plasmonic structures have shown only 10–30-fold increases in

nonlinear signal.^{31,32} Contributed by high-order nonlinear effects, the enhanced SHG was still observed in the electric-dipole-forbidden configuration at normal incidence. At weak pump intensities, these high-order SHG signals are negligible compared with dipolar contributions but are boosted significantly by the intense localized electric field gradients near the NPs. At off-normal angles, strong nonlinear responses can also be generated from propagating SLRs that exhibit angle-dependent resonance wavelengths along the $(\pm 1, 0)$ diffraction modes (Figure 2d). Therefore, the same NP array

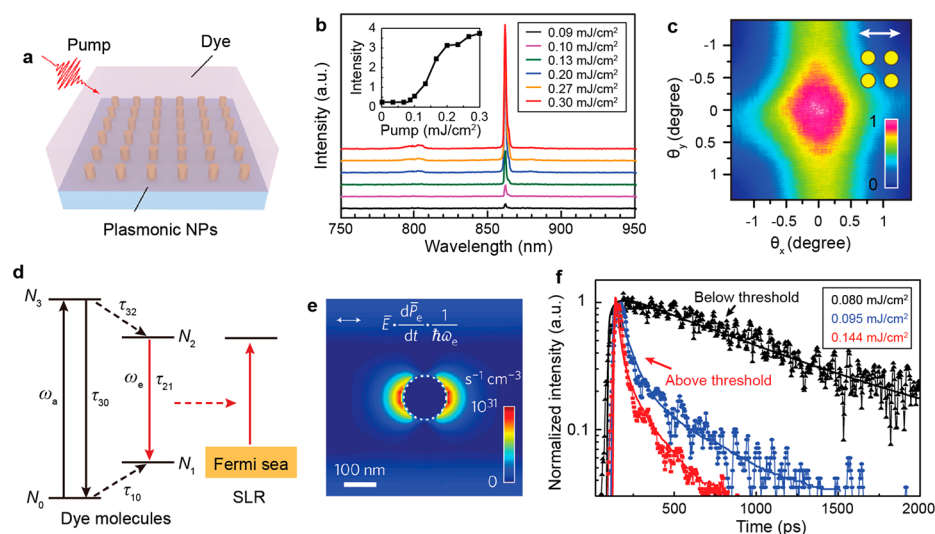


Figure 4. Nanoscale lasing from organic dye molecules integrated with square lattices. (a) Scheme of an SLR laser device. Adapted from ref 25. Copyright 2017 American Chemical Society. (b) Power-dependent lasing emission and (inset) input power–output intensity curves. Adapted from ref 20. Copyright 2015 Springer Nature. (c) Far-field beam profile with a highly directional lasing spot. Adapted from ref 20. Copyright 2015 Springer Nature. (d) Scheme of the process of energy transfer from dye molecules to an SLR mode in plasmonic lattices. ω_a and ω_e are the absorption and emission frequencies, respectively. (e) Calculated distribution maps of stimulated transition rate above threshold. Adapted with permission from ref 22. Copyright 2013 Springer Nature. (f) Decay times measured at pump intensities below and above threshold. Adapted from ref 20. Copyright 2015 Springer Nature.

can enhance SHGs over a range of pump wavelengths depending on the incident angle.

1.4. Tunable Hybrid Quadrupolar Lattice Modes in the Visible Regime

Conventional materials such as Au and Ag cannot support plasmon excitations at wavelengths shorter than those of their interband transitions because of large nonradiative losses.³⁶ In contrast, the plasmonic losses in metals such as Al are mainly radiative and can be mitigated by diffractive coupling. Therefore, Al can produce high-quality SLRs over the entire ultraviolet to near-infrared range,¹⁹ including across its interband transition wavelength ($\lambda \sim 800$ nm). With the same lattice geometry and particle diameter as Au lattices, Al NP lattices³⁷ can support SLRs with similar line widths at a slightly shorter wavelength ($\lambda = 839$ nm; Figure 3a). Charge density distributions calculated by FDTD simulations indicate that the SLR has in-plane dipolar character (Figure 3a, inset) with enhanced electromagnetic fields in the subwavelength vicinity of NPs.³⁷

Compared with in-plane SLRs, lattice coupling between out-of-plane surface plasmons is more tolerant to changes in lattice spacing. With increasing NP size, Al NP lattices can also sustain a sharp hybrid quadrupole lattice plasmon (HQL) resonance over the same wavelength range (Figure 3b). Calculated charge distributions on these large NPs show quadrupolar features that oscillate out of plane. These robust HQL modes can maintain a line width of <6 nm as the lattice spacing is stretched by 50% (Figure 3c). When arranged on a flexible substrate, the NPs can support reversibly tunable sharp resonances in the wavelength range $\lambda = 500$ – 600 nm under applied strain. We anticipate that these highly tunable, nanoscale-confined modes may provide unexplored prospects in plasmon-mediated nanocrystal growth³⁸ and photocatalysis.³⁹

2. RANDOMLY DISTRIBUTED EMITTERS COUPLED TO PLASMONIC LATTICES

The strong electromagnetic fields and large local density of optical states produced by SLRs can influence nanoscale chemical and physical processes.^{10,40} Interactions between SLRs and photoactive media have resulted in various phenomena, including hybridized polariton states in the strong coupling regime and enhanced absorption and emission in the weak coupling regime.⁴¹ Organic dye molecules are attractive emitters for investigating exciton–plasmon energy transfer because of their miscibility in polymers and solubility in solvents as well as diverse emission and absorption wavelengths. In this section, we discuss characteristics of nanoscale lasing from randomly distributed dye molecules combined with plasmonic NP cavities. We also describe how lasing wavelengths can be tuned dynamically by varying the refractive index environment and lattice spacing.

2.1. Nanoscale Lasing: Dye Molecules Integrated with Plasmonic Lattices

Lattice plasmon lasers consist of plasmonic NP lattices and dye molecules dispersed in a polymer matrix²² or an organic solvent^{20,42} (Figure 4a). As the device is optically pumped at room temperature, a lasing emission peak emerges above a threshold at which the input power–output intensity curves show a dramatic change in slope (Figure 4b). Unlike most plasmonic laser designs that lack beam directionality,^{43–45} square lattices of single-periodicity NPs integrated with organic dyes facilitate directional emission normal to the surface with a low divergence angle ($<1.5^\circ$) (Figure 4c), consistent with the band-edge mode at $k_{\parallel} = 0$ (Figure 2d) providing optical feedback. Recent work has shown smaller divergence angles ($<0.5^\circ$) at the onset of lasing.^{42,46} Exciton–plasmon interactions can be simulated using a semiquantum time-domain approach^{22,47,48} in which the electronic energy of the dye molecules is treated as a four-level quantum system and SLR excitations are described by Maxwell's equations (Figure 4d). When the molecules are excited by an external pump,

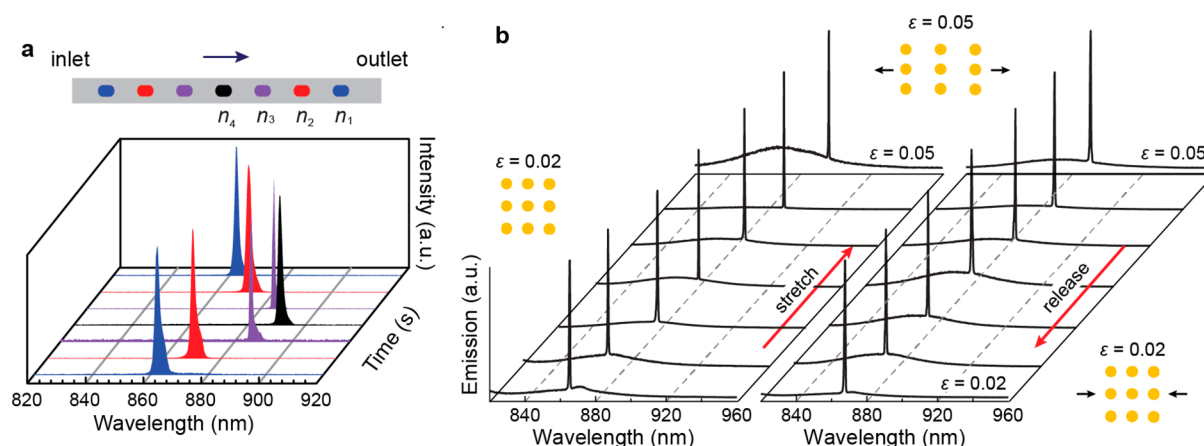


Figure 5. Tunable nanolasing by changing the refractive index of the liquid dye solution or varying the lattice spacing. (a) Dynamic laser with a microfluidic channel. Adapted from ref 20. Copyright 2015 Springer Nature. (b) Reversible tunable nanolasing under strain $\epsilon = 0.02$ – 0.05 . Adapted from ref 21. Copyright 2018 American Chemical Society.

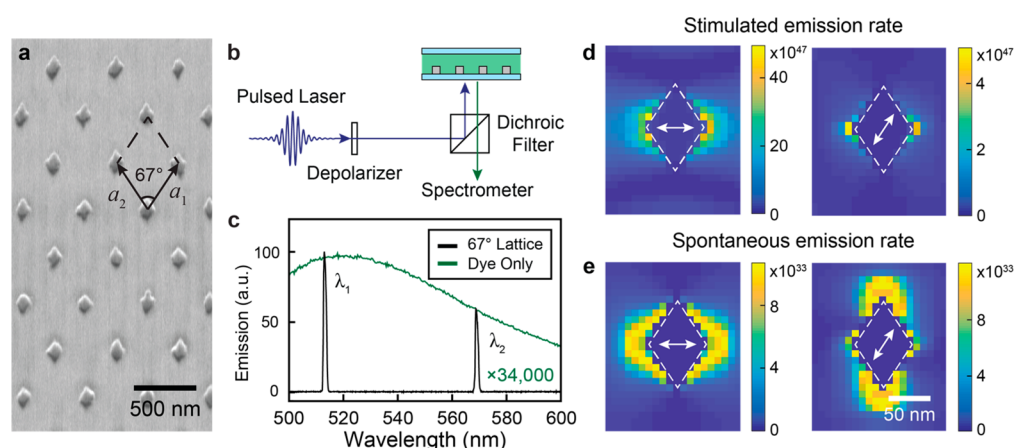


Figure 6. Low-symmetry nanoparticle lattices with polarization-dependent lasing and their simulated emission profiles. (a) SEM image of a low-symmetry NP array. (b) Scheme of the experimental setup. (c) Lasing spectrum for a lattice angle $\theta = 67^\circ$. The concentration of the dye C-481 was 20 mM. The photoluminescence spectrum was obtained below the lasing threshold and the lasing spectrum above the threshold. (d, e) Accumulated stimulated emission at the same hot spots regardless of the pump polarization. Adapted from ref 46. Copyright 2019 American Chemical Society.

population inversion builds up, and the excitons can resonantly transfer their energy to SLRs and result in lasing emission. This energy transfer process can be modeled by solving the net macroscopic polarization equation for the emission and absorption transition, the rate equations for electron populations, and Maxwell's equations.⁴⁹ A spatial map of the stimulated emission rate above threshold reveals that only dye molecules in the hot-spot regions around the NPs effectively participate in lasing action (Figure 4e), which confirms that highly localized electric fields are critical to the nanolasing mechanism. From the Purcell effect, the emission rates of an emitter increase within an optical cavity depending on the mode volume and quality factor.⁵⁰ SLR cavities also exhibit ultrafast lasing dynamics^{20,22,42} (Figure 4f). Below threshold, spontaneous emission dominates with a single exponential decay time close to the intrinsic lifetime of the dye molecules. As the pump intensity increases and approaches threshold, stimulated emission starts to dominate, and the decay lifetime shows a dramatic reduction from 715 to 20 ps.

2.2. Tunable Nanoscale Lasing by Changing the Refractive Index

Real-time tunable lasing emission can readily be achieved with SLR nanocavities because their resonance wavelengths are sensitive to changes in refractive index (n).^{10,20} Tunable lasing can be demonstrated by integrating a plasmonic NP lattice within a microfluidic channel such that liquid gain with different n can be introduced to the same system. By changing n , the lasing wavelength can be shifted over the entire emission bandwidth of the dye with no hysteresis (Figure 5a). In contrast to static orientations in a solid-state matrix, dye molecules show Brownian motion in a solvent; this property combined with pressure-driven flow inside a microfluidic channel can result in a constant population of active (albeit different) emitters in the plasmonic hot spots. Therefore, the lifetime of the nanoscale lasing device can be significantly extended and avoid photo-oxidation and photobleaching of dye molecules in a fixed host matrix.^{51,52} Notably, the wavelength tunability is enabled by tolerance of the SLR modes to minor mismatches between the refractive indices of the liquid superstrate and the solid substrate ($\Delta n < 0.05$).⁵³

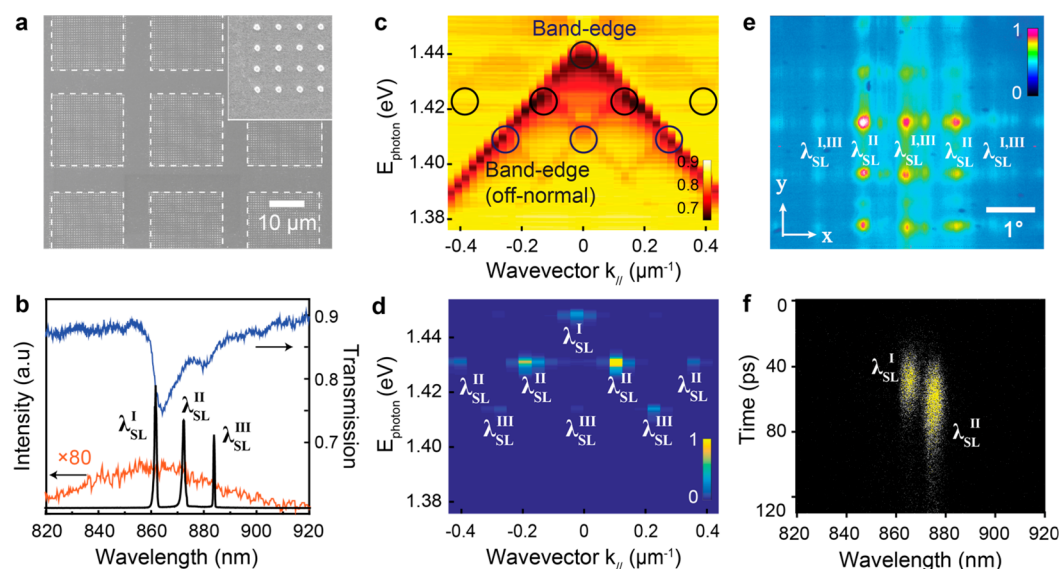


Figure 7. Controlled multimodal nanolasing from plasmonic NP superlattices. (a) SEM image of Au superlattice arrays with multiple length scales. (b) Lasing emission profile (in black), dye emission profile (in orange), and linear optical properties (in blue). The concentration of the dye IR-140 was 0.6 mM. (c) Measured optical band structure of superlattice arrays. (d) Measured angle-resolved emission of multimodal lasing. (e) Far-field beam profile. (f) Distinct ultrafast dynamics of multiple lasing peaks. Adapted with permission from ref 42. Copyright 2017 Springer Nature.

2.3. Stretchable Nanoscale Lasing by Varying the Lattice Spacing

The lasing wavelength can also be tuned by changing the lattice periodicity of plasmonic NP lattices on an elastomeric slab surrounded by dye molecules.²¹ Stretching and releasing the substrate can modulate the SLR wavelength by changing the interparticle separation and allow real-time tuning of the lasing output. Under multiple cycles of stretching and release, the emission wavelength shows excellent recovery with nearly zero hysteresis (Figure 5b). A much higher figure of merit ($\Delta\lambda = 31$ nm for $\Delta\epsilon = 0.03$, 18 nm) can be realized in stretchable lasing from plasmonic lattices compared with photonic crystals⁵⁴ ($\Delta\lambda = 26$ nm for $\Delta\epsilon = 0.23$, 150 nm). Also, the liquid gain medium ensures that there will always be emitters within the plasmonic hot spots for both applied and released strain. The quality of the SLR mode is preserved under applied strain because these large NPs support HQLs that are robust toward changes in lattice spacing (Figure 3c).

3. RATIONALLY DESIGNED LATTICE GEOMETRIES

Compared with NP lattices with a single periodicity, different lattice geometries and unit cells can produce distinct electromagnetic field distributions near the NPs, which offers a way to control the lasing emission properties.^{10,12,55} This section describes how low-symmetry plasmonic NP lattices can produce polarization-dependent lasing responses and how multiscale plasmonic NP superlattices can support multiple band-edge states for controlled multimodal nanolasing.

3.1. Low-Symmetry Rhombohedral Lattices

Figure 6a shows rhombohedral lattices of anisotropic Al NPs, where the lattice angle θ can be tailored in the patterning process.⁴⁶ Low-symmetry Al NP lattices support two SLR modes from the coupling of dipolar LSPs to both first-order and second-order diffraction modes. When the lattice was integrated with the green laser dye C-481 and pumped with an unpolarized laser (fs-pulsed, 400 nm), ultranarrow lasing peaks were observed from both SLR modes (Figures 6b,c). The

reduced lattice and unit cell symmetry resulted in tunable and switchable lasing emission. Faster lasing emission dynamics were observed when the transition dipole moments of the excited gain medium were aligned with the oscillation direction of the selected lattice mode.⁴⁶

The anisotropic NP shape can facilitate control over the spatial overlap of electromagnetic hot spots from the two SLR modes and dye molecules. Regardless of the pump polarization direction, the spatial distributions of stimulated emission rates for both lasing modes are confined at the same physical locations along the short axis (Figure 6d), in contrast to the spontaneous emission rate distributions that follow the pump polarization (Figure 6e). Dye molecules within these regions can selectively couple to different SLR modes and show polarization-dependent lasing emission. Thus, plasmonic NP lattices integrated with a gain medium offer prospects to control the exciton–plasmon energy transfer by varying the NP shape and lattice symmetry.^{46,55}

3.2. Multiscale Plasmonic Superlattices

Multiscale plasmonic lattices⁴²—finite arrays of metal NPs grouped into microscale arrays—provide an entirely different architecture to manipulate multiple band-edge states within a single substrate (Figure 7a). Designed using multiscale nanofabrication tools,²⁶ plasmon superlattices can be fabricated over square centimeter areas with independent control over the microscale features and interparticle spacings. The surprisingly long-range coupling (tens of microns) in plasmonic NP superlattices produces multiple sharp resonances that originate from coupling of the single-patch plasmon resonance to the high-order Bragg modes defined by the microscale patch periodicity.¹⁸ Integrated with liquid dye molecules, plasmonic superlattices can show multimode lasing action with multiple sharp lasing peaks and narrow spectral line widths (fwhm < 0.5 nm) (Figure 7b).

Compared with single-lattice NP arrays with only one band-edge state, the measured optical band structure reveals that plasmonic superlattices can support multiple band-edge modes at both zero and nonzero wavevectors (Figure 7c). A side-by-

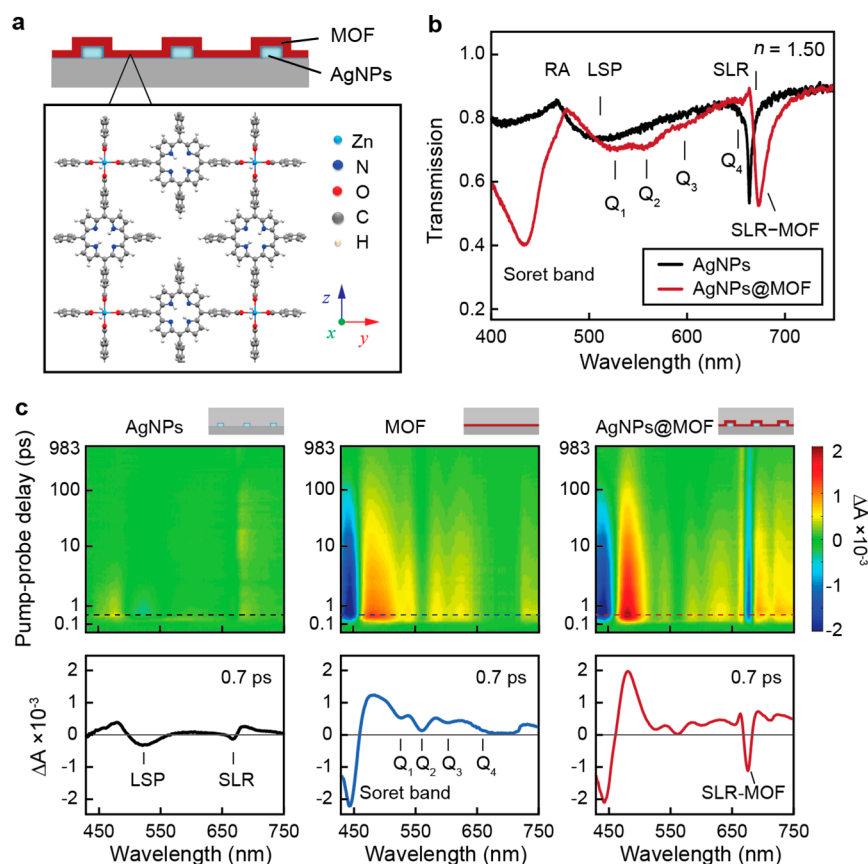


Figure 8. Spatially defined molecular emitters coupled to plasmonic nanoparticles. (a) Scheme of the MOF coated on Ag NP lattices and the crystal structure of Zn-porphyrin. (b) Coupling of the SLR in Ag NP lattices with Q_4 -band excitons in Zn-porphyrin. (c) Transient absorption provides evidence of electron population of the mixed plasmon–exciton state. Adapted from ref 24. Copyright 2016 National Academy of Sciences.

side comparison of an angle-resolved lasing emission map indicates that superlattice lasing modes have a one-to-one correspondence with the band edges (Figure 7d). Besides the central and single spot supported by single-lattice arrays,²² multiple satellite spots are observed, which are attributed to lasing modes at off-normal emission angles (Figure 7e). Superlattice plasmons can achieve accurate control over the wavelength and spectral separation of multiple cavity modes. The different electric near-field distributions and mode quality of the superlattice plasmons result in distinct ultrafast decay dynamics of the different lasing modes (Figure 7f). For symmetry-broken superlattice line patches, dynamical switching of lasing between a single mode and multiple modes can be realized simply by varying the lattice orientation with respect to the pump light polarization direction.⁴²

4. SPATIALLY ORGANIZED SOLID-STATE PHOTOACTIVE MATERIALS

Spatially organized photoactive emitters near plasmonic nanostructures can show enhanced emission and avoid unfavorable distance-dependent effects.^{56,57} In this section, we describe how crystalline materials such as metal–organic frameworks (MOFs) and hexagonal boron nitride (hBN) combined with plasmonic lattices can support exciton–plasmon mixed states and enhanced single-photon emission, respectively.

4.1. Precisely Positioned Molecular Emitters

MOFs are molecular crystalline structures consisting of organic ligands arranged between metal ion nodes; chromophore-

based ligands can function as molecular emitters. Figure 8a depicts a scheme to spatially organize chromophores via a MOF template on plasmonic NP lattices.²⁴ Luminescent Zn-porphyrin MOF thin films with the photoactive porphyrin derivative [5,15-diphenyl-10,20-bis(4-carboxyphenyl)-porphyrin] as the organic ligand were synthesized on Ag NPs using a layer-by-layer method. An SLR–exciton mixed state is produced because of coupling between the SLR in Ag NPs and Q_4 -band excitons in the MOF (Figure 8b). The Q -band excitons can be enhanced within plasmonic hot spots despite their weak oscillator strengths.⁵⁸ Femtosecond transient absorption measurements reveal excited-state interactions of the mixed modes (Figures 8c). For the hybrid AgNPs@MOF, a strong bleaching signal at the SLR-MOF resonance was observed, with absorption that persisted over a long time window (1 ns) without decaying to zero. Strikingly, the mixed SLR–exciton state exhibited the sharp line width of the lattice plasmon resonance and the long lifetime of porphyrin excitons. Compared with the pristine Zn-porphyrin MOF, hybrid AgNPs@MOF can achieve a large (16-fold) enhancement in photoluminescence intensity at wavelengths determined by the lattice modes.

4.2. Deterministically Coupled Quantum Emitters

Plasmonic lattices can also be integrated with 2D atomically thin semiconducting materials to modify the emission properties of quantum emitters. Isolated defects in hBN can behave as room-temperature single-photon emitters and be deterministically coupled to SLRs.⁵⁹ Before hBN is integrated with the plasmonic lattice, a single-photon emitter at a

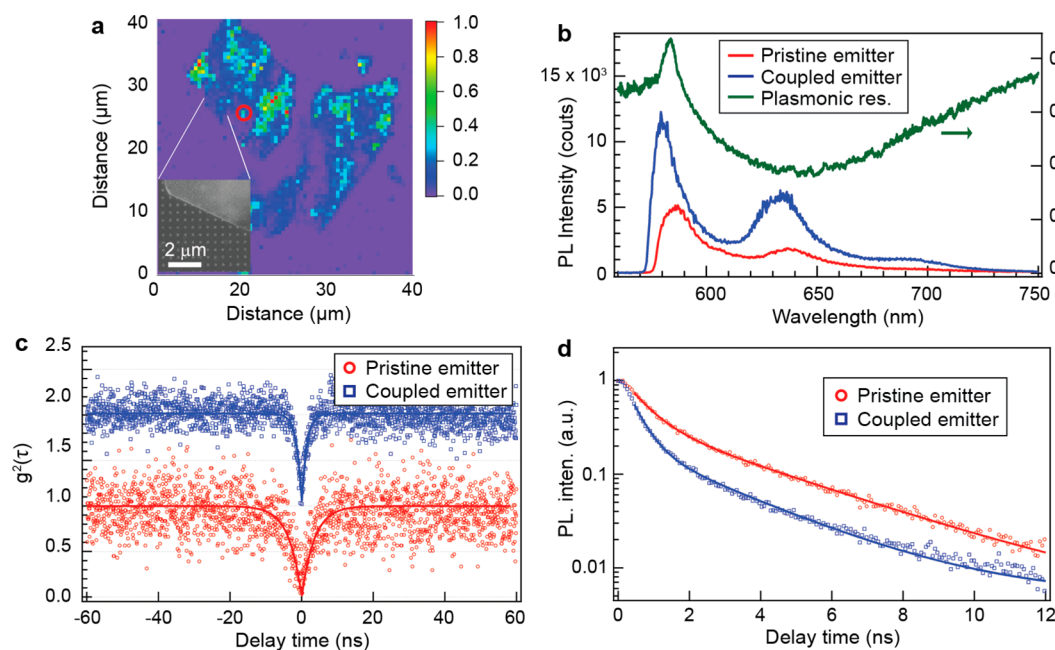


Figure 9. Coupling of quantum emitters in 2D materials to plasmonic nanocavity arrays. (a) PL confocal map of an hBN flake containing a single-photon emitter (red circle). The inset shows an SEM image of the flake on a Au NP lattice. (b) PL spectra of the pristine and coupled single-photon emitter and the transmission spectrum of the plasmonic lattice. (c) The quantum nature is maintained after coupling in the second-order correlation function $g^2(\tau)$. (d) Time-resolved PL measurements from the pristine and coupled systems. Adapted from ref 59. Copyright 2017 American Chemical Society.

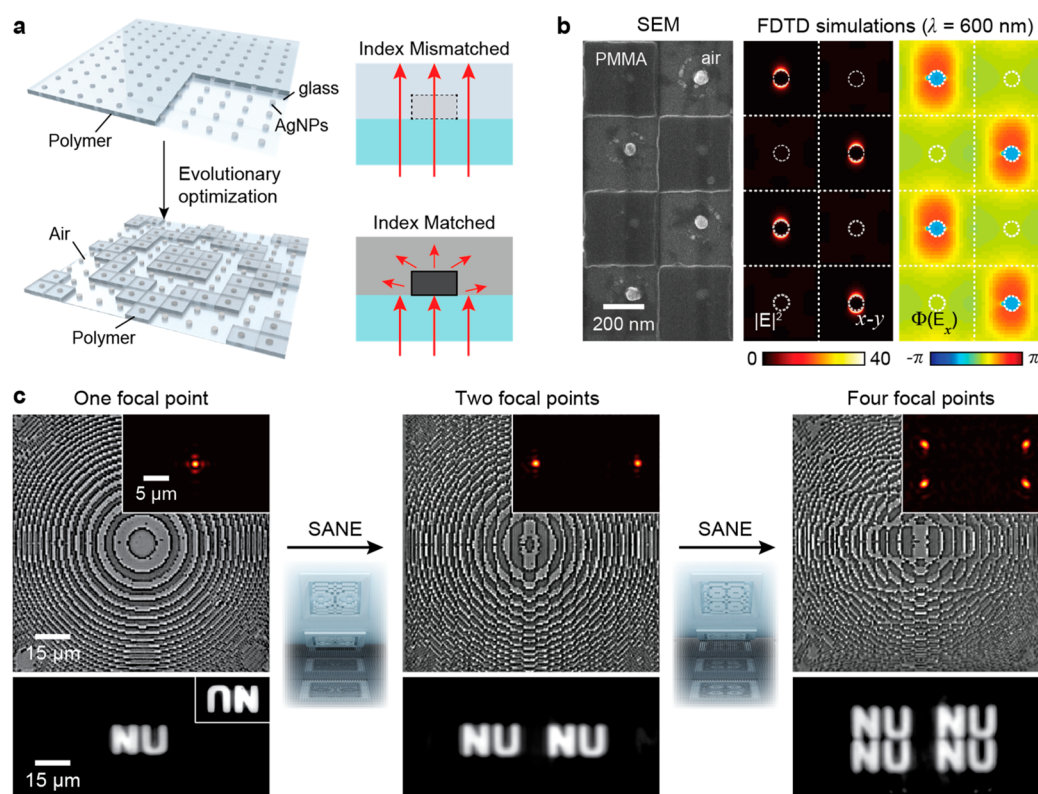


Figure 10. Lattice resonances enable reconfigurable metalenses for imaging. (a) Schemes depicting phase element design based on tuning of SLR by varying the dielectric environment. (b) (left) SEM images and near-field (middle) intensity profile, and (right) phase map for PMMA (refractive index $n = 1.47$) patterns on an NP array ($a_0 = 400$ nm). (c) (top) Optical microscopy images of SLR metalenses ($\lambda = 600$ nm) with reconfigurable SU-8 superstrate (MicroChem 2010, $n = 1.59$) with one, two, and four focal points and (bottom) the corresponding images formed from an object "NU" patterned on a Cr film. Adapted from ref 65. Copyright 2019 American Chemical Society.

frequency matching the SLR is first identified and characterized by PL imaging. Then the hBN flakes are transferred and

positioned on the NP lattice with a polymer membrane that can ensure a uniform refractive index environment (Figure 9a).

The emitter emission is enhanced by a factor of ~ 2.5 after the emitter is transferred onto the Au NP lattice (Figure 9b). The second-order correlation function $g^{(2)}(\tau)$ from the coupled emitters shows a dip at zero delay time below 0.5, which indicates that the single-photon emission preserves its quantum profile even after interacting with SLRs (Figure 9c). In addition, the coupled emitters exhibit enhanced emission rates and reduced fluorescence lifetimes (Figure 9d), consistent with Purcell enhancement in the weak coupling regime.

5. EMERGING APPLICATIONS AND FUTURE OUTLOOK

5.1. Lattice-Resonance Metalenses for Reconfigurable Imaging

Besides electric field enhancement and light confinement, plasmonic NP lattices can modulate the wavefront phase profiles that govern light propagation. Conventional metalenses—planar lensing elements—focus light by shaping the optical wavefront using subwavelength phase units^{60–62} and have been used in compact imaging devices.^{63,64} Recently, light focusing and imaging by SLR excitations have been demonstrated using metalenses based on coupled plasmonic NPs.⁶⁵ This architecture is based on two types of elements: (1) Ag NPs exposed to air and (2) Ag NPs covered by a thin dielectric block (Figure 10a). These building blocks can be arranged on NP arrays using poly(methyl methacrylate) (PMMA) as the dielectric film to realize SLR metalenses (Figure 10b). Significant wavefront changes result only on NPs covered by PMMA that matches the refractive index of the glass substrate. SLR excitations are maintained even when a fraction of the NPs have mismatched surroundings. Therefore, both the amplitude and phase of light can be tuned at the single-particle level by patterning the superstrate dielectrics on the lattice. Designed by a lattice evolution algorithm,^{66,67} multifocal SLR metalenses can realize concurrent image formation in two focal planes separated by 180 μm .⁶⁵

Metalenses can be reconfigured by reshaping the dielectric blocks on the NPs in a solvent-assisted nanoscale embossing process.⁶⁸ Polydimethylsiloxane (PDMS) masks molded from patterned silicon templates with optimized metalens designs can erase existing patterns with solvent and define new features in a single embossing step. Reversible switching between lens configurations is possible using multiple PDMS masks that have metalens designs for different focal points. Adaptive imaging has been demonstrated on this SLR platform by sequentially evolving the superstrate dielectrics from single-focus to two-foci and four-foci designs during the same imaging session to obtain a single image or multiple images (Figure 10c).

5.2. Future Prospects

Plasmonic NP lattices can now function as a platform material for a wide range of applications. They can easily be combined with emerging materials ranging from other 2D materials to quantum emitters to upconverting nanoparticles. For example, robust solid-state nanolasers may be realized by integrating SLRs with colloidal quantum dots that exhibit extraordinary quantum yield.⁶⁹ Such nanocavities also have the potential to enhance upconversion efficiencies from upconverting nanoparticles,⁷⁰ whose long lifetimes could facilitate population-inversion buildup for nanolasing under continuous-wave pumping. In addition, SLRs may open up the possibility to

modulate chemical reactions by strong coupling to the vibrational modes of molecules.⁷¹ Tunable stereoselective reactions are also promising using chiral plasmonic lattices with engineered symmetry-breaking unit designs. Finally, plasmonic NP lattices offer prospects to test the fundamental question of whether topological edge states⁷² can exist in lossy systems given that SLRs strongly suppress radiative losses. These topologically protected states have only been demonstrated in low-loss dielectric and magnetic materials, and the effect of losses on such exotic properties is unknown.⁷³ Looking forward, we anticipate even more opportunities for plasmonic NP lattices to manipulate nanoscale processes in active and functional materials and that they will play a key role in driving enhanced photocatalysis, nanoscale bioimaging, compact optoelectronics, and light-sensitive chemical reactions.

AUTHOR INFORMATION

Corresponding Author

*E-mail: todom@northwestern.edu.

ORCID

Danqing Wang: 0000-0002-7369-1944

Jun Guan: 0000-0001-8667-1611

Jingtian Hu: 0000-0002-0528-6250

Teri W. Odom: 0000-0002-8490-292X

Notes

The authors declare no competing financial interest.

Biographies

Danqing Wang received her B.S. degree in Physics at Nanjing University (Nanjing, China) in 2013. She received her Ph.D. degree in Applied Physics at Northwestern University in 2019, coadvised by Prof. Teri W. Odom and Prof. George C. Schatz. Her research focuses on the structural engineering of plasmonic nanocavities and their light–matter interactions with different gain materials.

Jun Guan is a Ph.D. candidate in the Applied Physics Program at Northwestern University, coadvised by Prof. Teri W. Odom and Prof. George C. Schatz. She received her B.S. degree in Physics at Sichuan University (Chengdu, China). Her research focuses on integrating plasmonic lattices with quantum emitters and engineering the light–matter interactions at the nanoscale.

Jingtian Hu received his B.S. degree in Materials Science and Engineering from the University of Illinois at Urbana–Champaign in 2013. In 2019, he received his Ph.D. degree in Materials Science and Engineering at Northwestern University, advised by Prof. Teri W. Odom. His research focuses on predictive design of plasmonic metasurfaces by evolutionary algorithms.

Marc R. Bourgeois is a Ph.D. candidate in the Department of Chemistry at Northwestern University. He received his B.A. degree from Northwestern University in 2012 and his M.S. degree in Physics from the University of Arizona in 2014. He is currently working under the guidance of Prof. George C. Schatz, investigating the optical properties of plasmonic nanomaterials.

Teri W. Odom is the Charles E. and Emma H. Morrison Professor of Chemistry and Chair of the Department of Chemistry at Northwestern University. She obtained her B.S. degree in Chemistry at Stanford University and her Ph.D. in Chemical Physics at Harvard University. Her research focuses on designing structured multiscale nanoscale materials that exhibit extraordinary size- and shape-dependent optical and physical properties.

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